D4.1 Natural selection (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 **DEFINE** gene pool. [D4.1.9] [1]

2 **DEFINE** allele frequency. [D4.1.9] [1]

3 **STATE** the two Hardy-Weinberg equations. [D4.1.13] [2]

4 **NAME** the type of selection that favours both extremes of a trait. [D4.1.12] [1]

5 **STATE** two conditions needed for Hardy-Weinberg equilibrium. [D4.1.14] [2]

6 **STATE** one example of artificial selection. [D4.1.15] [1]

Section B: Understanding

14 MARKS

7 **EXPLAIN** how natural selection changes allele frequencies in a gene pool. [D4.1.11] [3]

8 **DISTINGUISH** between stabilizing and directional selection. [D4.1.12] [3]

9 **EXPLAIN** why populations that are separated geographically can have different allele frequencies. [3]
[D4.1.10]

10 **EXPLAIN** why the evolution of antibiotic resistance is natural, not artificial, selection. [D4.1.15] [2]

11 **EXPLAIN** what it shows if the genotype frequencies of a population do not fit the Hardy-Weinberg equation. [3]
[D4.1.14]

Section C: Working with evidence

12 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [D4.1.13, D4.1.14]

In a species of snail, the allele for an unbanded shell (B) is dominant to the allele for a banded shell (b). In a woodland, 1000 snails were sampled: 160 were banded. Ten years later, after thrushes began feeding there, 1000 snails were sampled again: 90 were banded.

a **CALCULATE** the frequency of allele b in the first sample. [2]

b **CALCULATE** the frequency of allele B in the first sample. [1]

c **CALCULATE** the number of heterozygous snails expected in the first sample. [2]

d **CALCULATE** the frequency of allele b in the second sample. [2]

e **SUGGEST** an explanation for the change in allele frequency. [2]

f **IDENTIFY** the type of selection shown. [1]

g **IDENTIFY** which Hardy-Weinberg condition was not met, and give a reason. [2]

Section D: Longer answer

6 MARKS

13 **COMPARE** natural selection and artificial selection. [D4.1.15] [6]